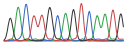
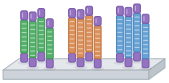


Nanopore



Sanger



Illumina



samples



AID-null controls

same sequencing run

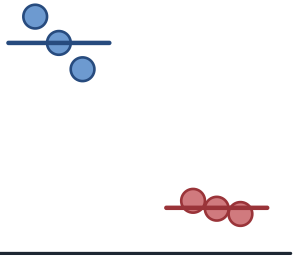
SHMtrace

reads: each candidate mutation tested against the errors of matched AID-null libraries

Sanger: quality-filtered counts; background reported

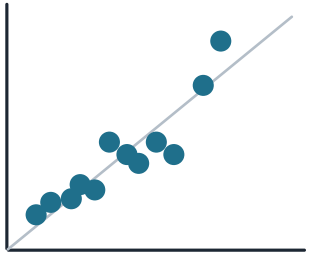
groups compared animal by animal

Low background



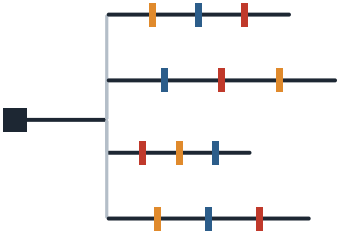
1–4 calls per control set

Pooled spectra



pooled substitution spectrum, nanopore vs Sanger, r = 0.92

Clones and trees



read-level reconstruction from full-length reads